

A 250-MHz 18-Mb Full Ternary CAM With Low-Voltage Matchline Sensing Scheme in 65-nm CMOS

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Abstract—An 18-Mb full ternary CAM with low-voltage matchline sensing scheme (LVMLSS) is designed and fabricated in 65-nm bulk CMOS process. LVMLSS has three key techniques: voltage down converter, differential sense amplifier with matchline isolation, and reference voltage generation scheme. With these techniques, LVMLSS can reduce the dynamic power consumption of matchlines to 33% compared with conventional one and realizes 42% fast match-line sensing. At 1.0-V typical supply voltage, 250-MHz search frequency is achieved. The power consumption of fully paralleled search operation at 250 MHz is 9.3 W, which is 66% smaller than previous work. This work has realized high-speed, low-power, and robust large-scale TCAM. We believe that this work will greatly contribute to reducing the power of network systems.

Index Terms—Differential sense amplifier, low-voltage matchline, reference voltage generator, ternary content-addressable memory (TCAM), voltage down converter.

I. INTRODUCTION

FULLY paralleled ternary content-addressable memories (TCAMs) are very attractive devices for not only networking applications such as packet routers or switches but also database search applications [1]–[5]. However, it is challenging work to realize both high-density (= high-capacity) and high-speed chip due to large power consumption and less signal integrity such as large power-ground noise at search operation. Process scaling is important to achieve high-density and high-performance TCAM, but it incurs the decrease of operating margin caused by threshold voltage variation under sub-65-nm process.

Fig. 1 shows a conceptual view of a TCAM. The search data is compared with every entry simultaneously. The number of bits in an entry is usually large, ranging from 36 to 144 bits. A typical TCAM employs a table size ranging from a few hundred entries to several thousand entries. In conventional precharge-high matchline (ML) sensing scheme [1], all MLs are precharged to high (= core supply voltage), and then if all bits on a ML match

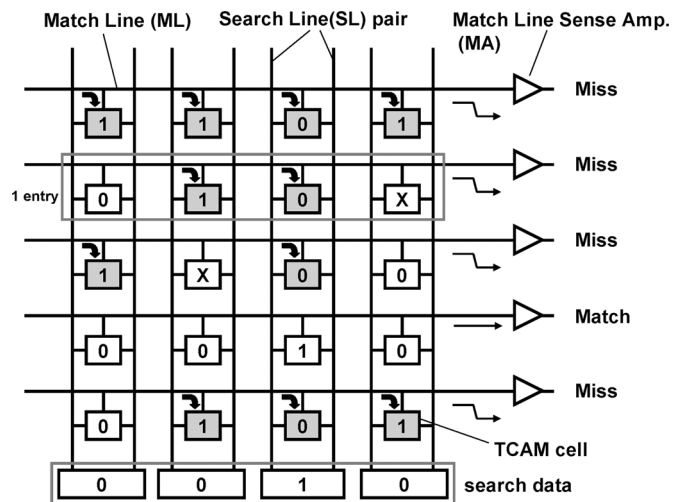


Fig. 1. Conceptual view of TCAM.

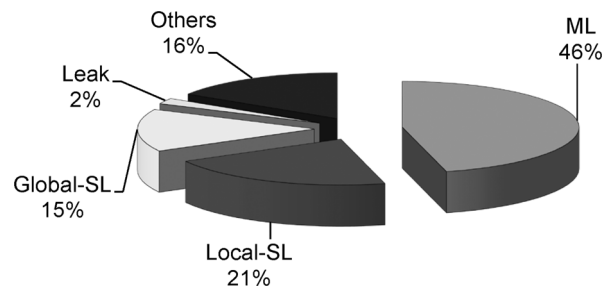


Fig. 2. Power consumption analysis of our previous work.

with search data (= match), the ML will remain high, otherwise (= miss), the ML is discharged. Typically, there are only a small number of matches. Analysis for conventional work shows that the power consumption of MLs occupies nearly a half of the total power consumption of TCAM as shown in Fig. 2. It indicates that power reduction of MLs is the most effective way to achieve low power operation.

Although many architectural, algorithmic, and circuit-level studies have been reported, the effectiveness of architectural and algorithmic schemes often depends on search data patterns. Among them, one straightforward method of reducing the ML power consumption is to reduce the ML voltage swing. This method, however, requires further improvements of sensing scheme to overcome process variation and power-ground noise. Various small swing ML sensing circuits have been reported. Most of them, however, use single-ended detect circuits and

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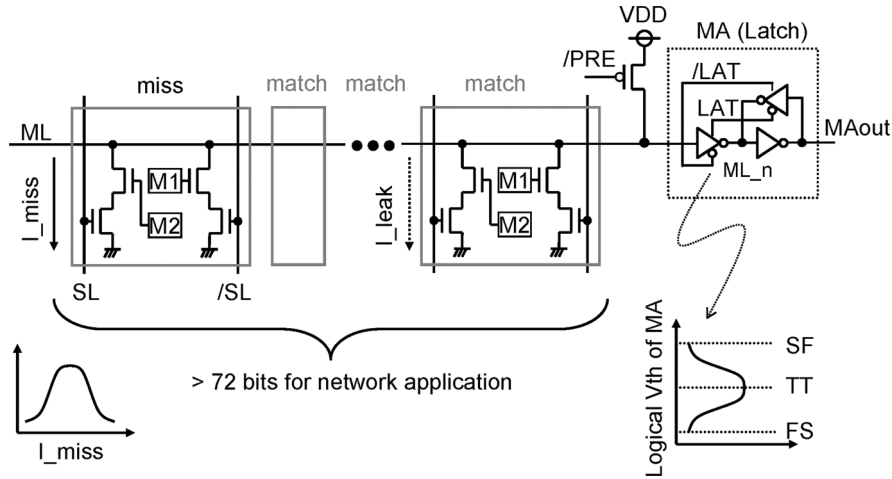


Fig. 3. Conventional circuit of one entry.

the logical thresholds will suffer from process variation in sub-65-nm processes.

In this paper, we propose low voltage matchline sensing scheme (LVMLSS) with small TCAM bitcell in 65 nm process. The proposed LVMLSS has lower power and higher speed operation than conventional precharge-high scheme, keeping robustness under 65 nm process.

The composition of this paper is as follows. In Section II, conventional matchline sensing scheme is introduced. In Section III, a low voltage matchline sensing scheme is proposed. In Section IV, the architecture and bitcell of TCAM are described. In Section V, simulation and measurement results are shown. A conclusion is described in Section VI.

II. CONVENTIONAL MATCHLINE SENSING SCHEME

Fig. 3 shows a conventional circuit of one entry. The match amplifier (MA) consists of clocked inverter and latch. The operation of this circuit consists of two phases. In the first phase, ML is precharged to VDD. In the second phase, ML is discharged by search transistor if there is at least one “miss” cell. Then, ML level is compared with logical threshold (V_{th}) of clocked inverter and MA indicates the results as match or miss.

To achieve high speed, low power and large capacity TCAM, there are two challenges in this circuit. The first one is for logical threshold voltage variation due to global process variation. If it becomes low, the sensing speed becomes slow, and if it becomes high, the circuit suffers from power-ground noise which is getting severe for large capacity TCAM.

The second one is for local process variation. To achieve large capacity, the bitcell should be small. But this will suffer from variation of discharge current I_{miss} due to local process variation.

Fig. 4 shows the simulation results of conventional MA circuit with two process conditions. In the simulated circuit, the V_{th} is set around half VDD. Considering global and local process variation, the speed of ML-discharge is so slow that MA can not detect the miss result correctly even after 4 ns from the rising edge of searchline (SL). To shorten the detecting time, setting V_{th} high is one way. However, it make sensing

operation more susceptible to VDD noise and ML droop by off-leak of “match” cells.

III. LOW-VOLTAGE MATCHLINE SENSING SCHEME (LVMLSS)

Fig. 5 shows the proposed LVMLSS schematic diagram. There are three key features in this scheme: 1) voltage down converter; 2) differential sense amplifier with matchline isolation; and 3) reference voltage generation scheme.

A. Voltage Down Converter (VDC)

The simplest conventional MA is clocked inverter. It consumes large power at search operation, because MLs are precharged to full VDD level and almost all MLs are discharged to ground in every search cycle. On the other hand, in the proposed LVMLSS, the ML precharge level is set to 0.3 V, which is generated from 1.0 V (= core supply voltage) by an on-chip VDC. The VDC is a conventional type used, for example, in DRAM. It consists of a reference level generator, voltage comparator, and driver. The reference level for the comparator is generated by resistor strings and is able to be trimmed. With this scheme, we can reduce the power consumption of MLs to about one third of conventional one.

B. Differential Sense Amplifier With Matchline Isolation

To make sense amplifier operate fast, we apply charge-captured cross-coupled sense amplifier. This sense amplifier has no offset in a pair of transistors. It is different from CIMDC scheme [3]. CIMDC is using the offset voltage between an input pair transistors of sense amplifier (SA) as a reference voltage. However, it will be affected by temperature. On the other hand, in our case, a pair of PMOS transistors at cross couple are the same size, and the size of the pair of NMOS transistors are the same. When the sense amplifier is activated, the sense node MLc and REFc are isolated by turning off matchline isolation gate (MLI), and the voltage levels of ML and the reference level of REF are captured into captured matchline node (MLc) and captured reference node (REFc).

The isolation of ML and REF from SA has an advantage also for high-speed operation of SA because the heavy load of ML impedes the SA operation unless it is isolated.

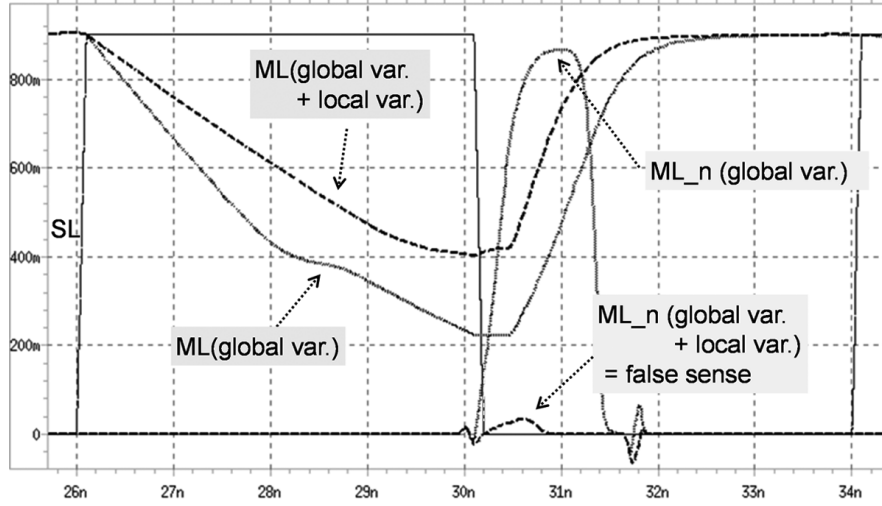


Fig. 4. Simulation waveform of conventional MA with global and local process variation.

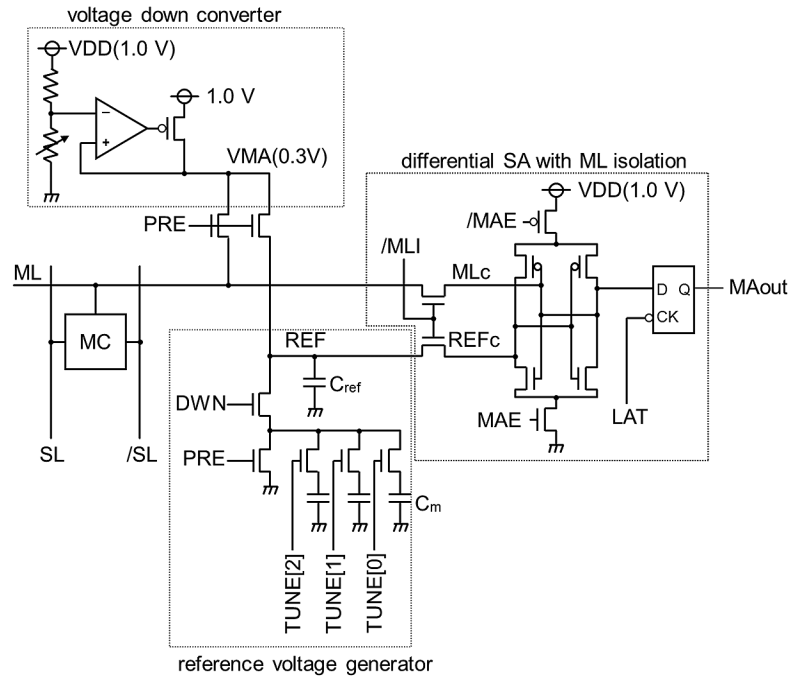


Fig. 5. Schematic of proposed LVMLSS.

The combination of VDC and differential sense amplifier with isolation is analogous to the sensing scheme of DRAM; however, noise immunity to search operation is different. In DRAM case, bitline pair is placed with almost minimum design rule. To avoid coupling effect from adjacent bitlines, some countermeasures such as twisted bitline layout are adopted. On the other hand, the space between adjacent MLs is wide and there is ground line in between. The coupling effect at search operation is negligible even though the precharge level is 0.3 V.

C. Reference Voltage Generation Scheme

Fig. 6 explains the ML reference voltage generation scheme of LVMLSS. Another important advantage of LVMLSS is that this scheme is insensitive to global process variation because the reference voltage is determined only by the ratio between C_{REF}

and C_{mi} , where C_{REF} is the capacitance of REF node and C_{mi} is the MOS capacitor. The generated voltage of REF node V_{REF} is expressed by the equation

$$V_{REF} = VMA \times C_{REF} / (C_{REF} + C_{mi}). \quad (1)$$

For the sake of mitigating offset effect of sense amplifier and timing degradation due to the variation of discharge current I_{miss} of the search transistor at the NMOS slow process corner, a concise tuning scheme is prepared. Fig. 7 shows the simulation results of V_{REF} by this scheme. The tuning range is designed to cover the global process variation of the search transistor. V_{REF} should be set higher at slow corner to counter the degradation of I_{miss} , and lower at fast corner to counter the effect of I_{leak} .

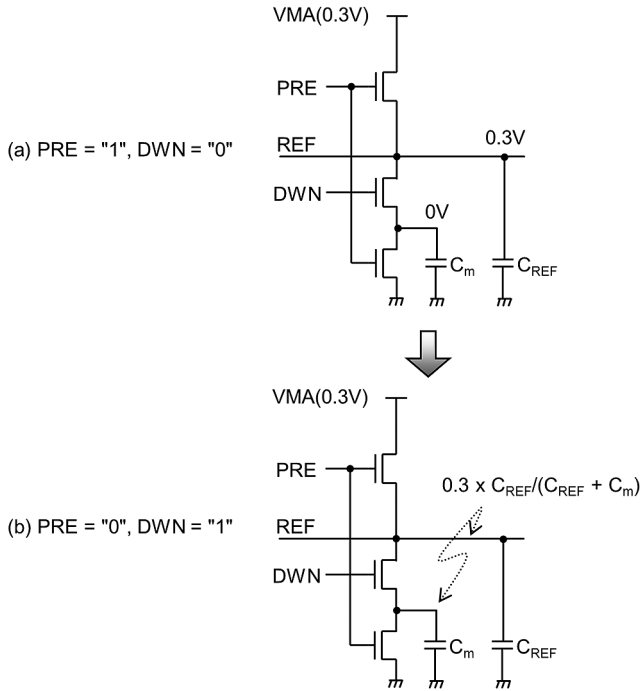
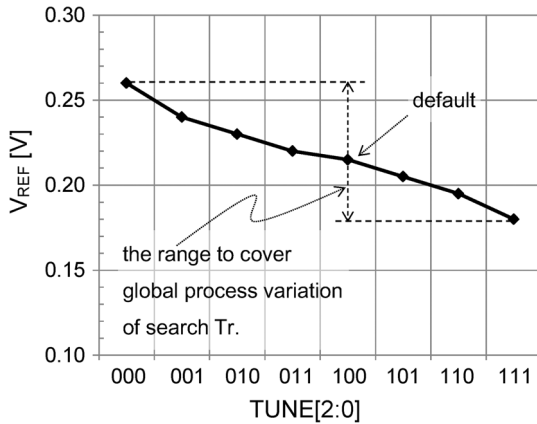


Fig. 6. ML reference voltage generation scheme.

Fig. 7. Simulation results of reference voltage V_{REF} tuning.

The tuning code is to be determined in testing phase so that the overall operation can meet the external specification on operational speed. The same code is fed for all MLs with fuse trimming.

In addition to high immunity to global process variation, the LVMLSS is less susceptible to the fluctuation of precharged level of MLs and ground noise because V_{REF} is generated in every search cycle.

D. Overall Operation

The timing diagram of the proposed LVMLSS is shown in Fig. 8. The rising edge of CLK turns off ML precharge signal (PRE), and asserts SL and DWN signal. Slightly after PRE turns off, DWN starts to generate V_{REF} by charge distribution within search period. Then, the voltage level of REF becomes approximately 0.2 V, which is lower than the precharged level of VMA.

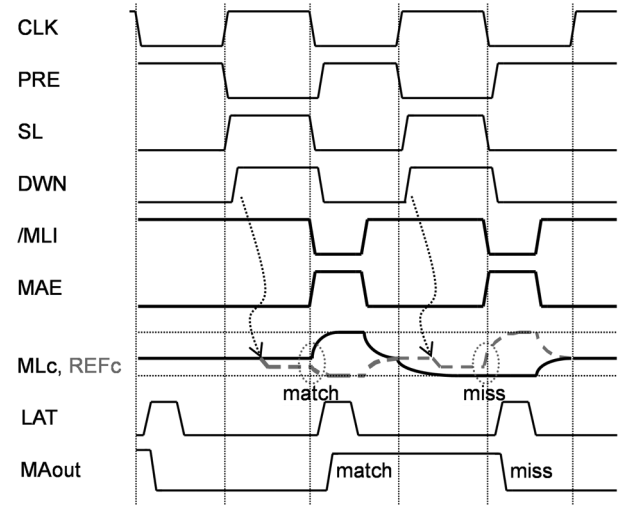


Fig. 8. Timing diagram of LVMLSS.

V_{REF} generation is executed in every cycle, therefore, even if there is low-frequency noise on the VMA, V_{REF} becomes lower than VMA. High-frequency power-ground noise can be eliminated by decouple capacitor.

When SLs are activated, if the stored data does not match with search data (it is the “miss” state), ML falls from VMA to VSS. In the case of the stored data matches, ML remains precharged level VMA.

The falling edge of CLK turns off matchline isolation (/MLI) and each voltage level of ML and REF at this moment is captured into MLc and REFc. At the same time, match amplifier enable signal (MAE) is activated to sense whether MLc level is higher or lower than REFc. Then, by activation of LAT signal, latch circuit takes in the result of sensing and outputs MAout. After the LAT signal activated, ML and REF are precharged to VMA level.

There is no timing overhead due to the reference voltage generation. Because the generation is executed in parallel with search operation (= ML discharge operation) and is faster than ML discharge speed, it can completely be seen as background operation. Fig. 9 shows the timing comparison of operational phase between conventional MA and LVMLSS. Search operation starts by asserting SL and DWN signals simultaneously. After enough duration for ML discharge, /MLI isolates SA from ML and MAE enables SA. The latch is opened by LAT slightly after MAE assertion. Sense and latch phase can be overlapped with ML precharge phase because ML and SA are isolated by the MLI transistor in this period.

In the conventional case, latch-open phase cannot be overlapped with the precharge phase because there is no isolation transistor between ML and MA. Likewise, it cannot be overlapped with the ML discharge phase because the ML node is intermediate voltage for long duration and it cause through-current in MA.

In this way, LVMLSS expenses no timing overhead, even though it needs additional signals (e.g., /MLI, DWN, and MAE) to conventional MA.

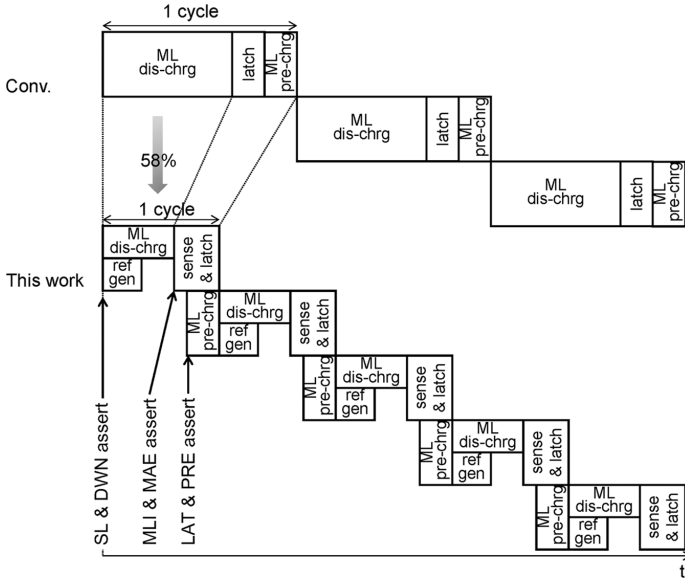


Fig. 9. Timing comparison between conventional MA and LVMLSS.

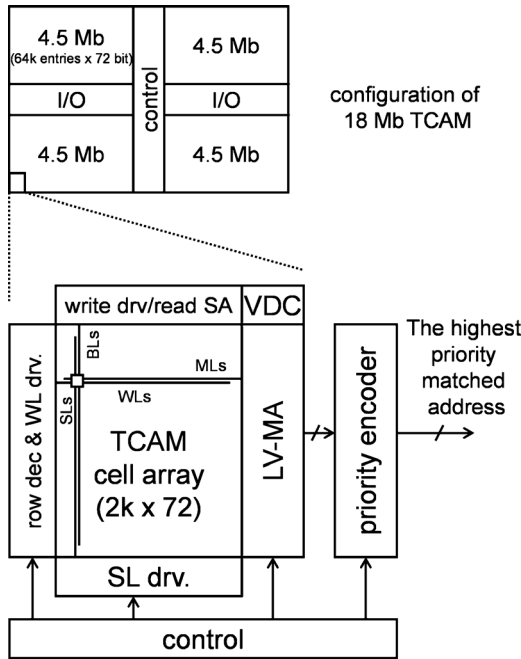


Fig. 10. Architecture of proposed TCAM.

IV. ARCHITECTURE AND TCAM BITCELL

Fig. 10 shows the architecture of proposed 18-Mb TCAM chip. It consists of four 4.5-Mb macros, and each macro consists of 2 k-entry $\times$ 72-bit blocks. Each block has row decoder, write driver, and sense amplifier for write and read of TCAM cells. For the search operation, SL drivers broadcast search data onto SLs, and each TCAM cell compares it with stored data. Search results come up to as a voltage level of MLs and low-voltage match amplifiers (LVMA) output sensed results to priority encoder. Priority encoder (PE) outputs the highest priority matched entry address.

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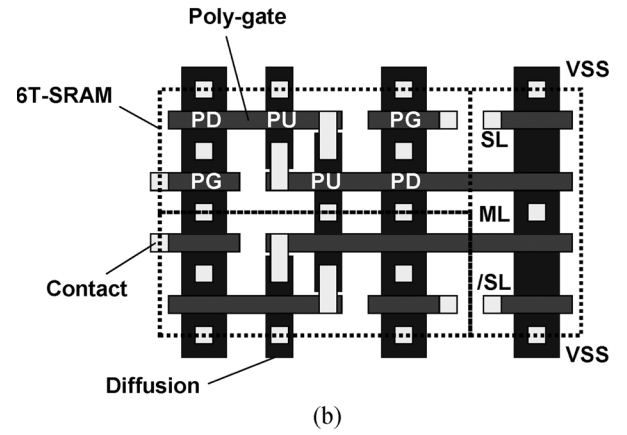
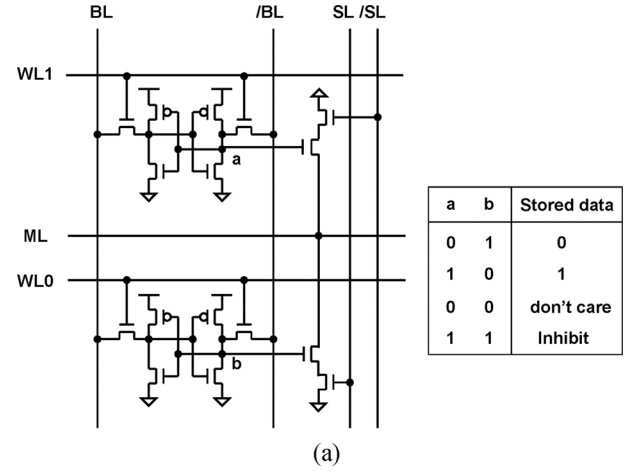


Fig. 11. Schematic and layout of proposed 16T TCAM bitcell. (a) Schematic. (b) Layout.

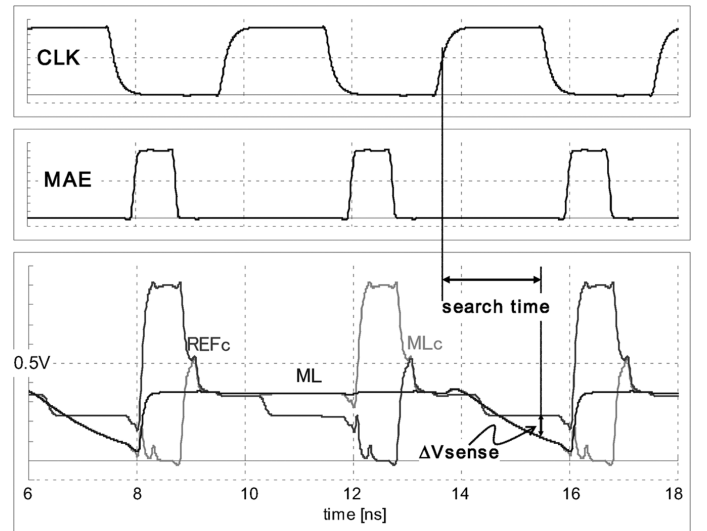


Fig. 12. Simulation results of the proposed LVMLSS.

A VDC module is placed in every 2 k-entry array for high-speed response to precharge MLs.

Fig. 11(a) shows the TCAM bitcell schematic. It is composed of two standard 6T-SRAM bitcell to store the three-state data and additional four NMOSs as a comparator. The transistors connected to SLs are placed opposite side of ML (=

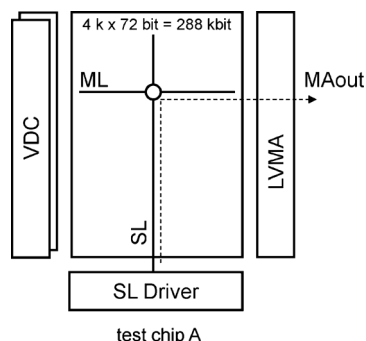


Fig. 13. Configuration of test chip A.

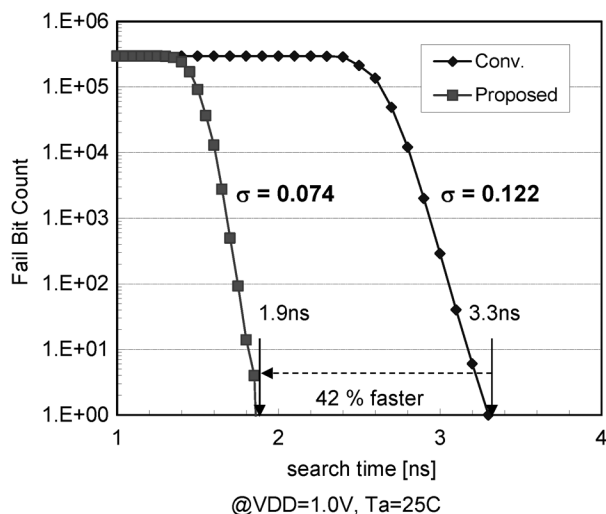


Fig. 14. Search time distribution of 4k entry x 72-b test chip A.

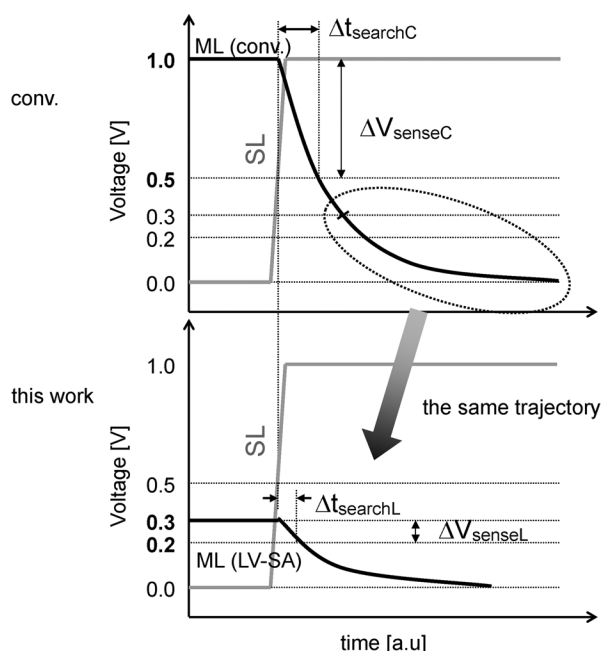


Fig. 15. Comparison of conventional and LVMLSS search times.

power-ground side) to avoid charge-sharing problem during matchline evaluation as explained in [1].

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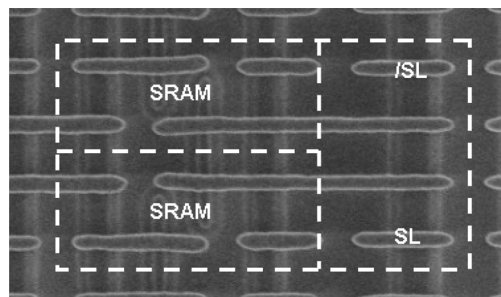


Fig. 16. Top SEM view of the TCAM bit cell after poly etching.

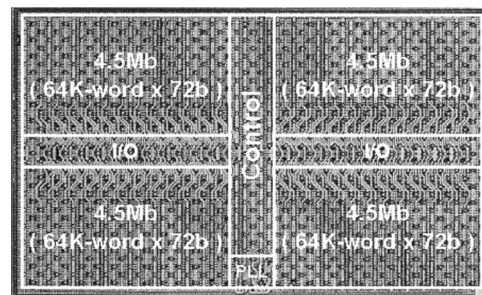


Fig. 17. Microphotograph 18 Mbit TCAM test chip B using 65-nm bulk CMOS technology.

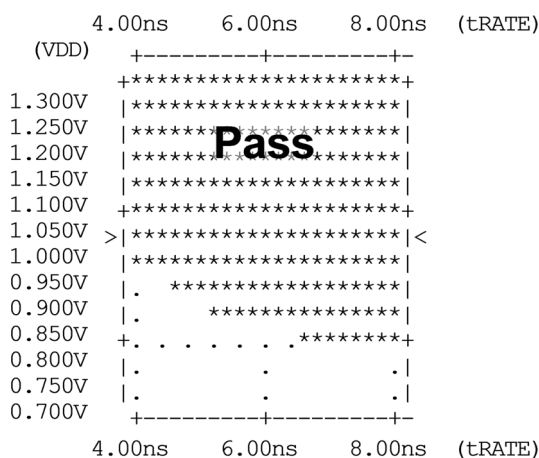


Fig. 18. Shmoo plot at the search operation of the 18 Mbit TCAM.

Fig. 11(b) illustrates the proposed front-end-of-line layout of TCAM bitcell (showing up to the contact layer). Two standard 6T-SRAM bitcell are combined with additional four NMOS. All poly-gates for PMOSs and NMOSs are arranged uni-direction with lithography friendship, achieving the high-density bitcell.

V. SIMULATION AND MEASUREMENT RESULTS

Fig. 12 shows the simulation results of the LVMLSS, corresponding to the timing diagram in Fig. 8. Two search cycles are shown in this simulation. The first search cycle is of “match” operation and the second one is of “miss” operation.

The ML reference signal REF_c is generated in every search cycles followed by confinement by lowering of MLI, and the voltage level of REF_c is around 0.2 V. The confined ML level, ML_c , is compared with REF_c at the activation of MAE.

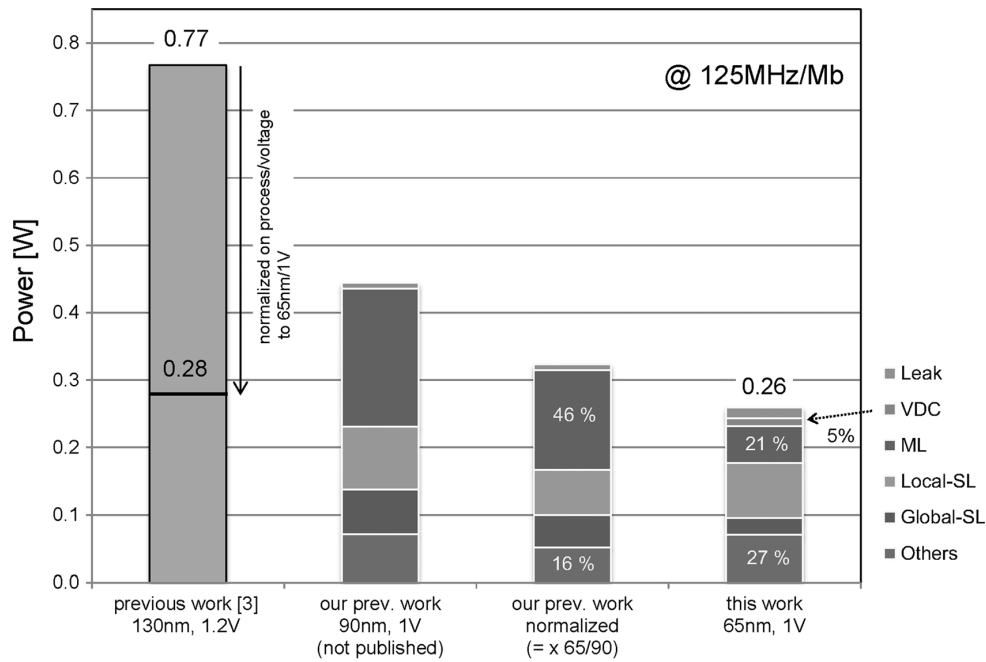


Fig. 19. Measured power of 18 Mbit TCAM and comparison of the constituents.

In sub-65 nm process, the small cell size leads to an enlarged variation of drain current and threshold voltage of transistors. It is important to overcome the local variation of cell transistors for high-density chip. To verify the LVMLSS search time and its variation with a small TCAM cell in 65 nm, we fabricated an array macro test chip A.

Fig. 13 shows the configuration of test chip A of a 288-kb array. The purpose of this chip is to measure the characteristics of search speed and its distribution; therefore, there is no priority encoder. Fig. 14 shows the measurement results of search time distribution at VDD = 1.0 V. The search time is defined as from the activation of SL to the time where the difference between REF_c and ML_c is larger than sense voltage ΔV_{sense} . The array test chip A includes 4 k-entry $\times$ 72-b arrays with the proposed LVMLSS and conventional MA. The measured search pattern is of one bit miss state; it is the worst case in search operation.

The worst search time of proposed LVMLSS is 1.9 ns, which is 42% faster than that of conventional one. This is because the LVMLSS can sense “miss” by only a 100-mV swing (ΔV_{senseL}) of ML, but a conventional one needs more than half of the VDD swing (ΔV_{senseC}) to detect “miss,” as explained in Fig. 15.

The distribution of access time is 0.074 ns, which is 39% smaller than that of the conventional one. This means that the proposed LVMLSS has more margins for the variation of drain current of small cell transistor than conventional one.

To set the voltage swing of the conventional scheme 100 mV is possible by setting the logical V_{th} of inverter-type MA high. However, in this case, it becomes more susceptible to the imbalance of on-resistances between PMOS and NMOS by global process variation, the local variations of PMOS and NMOS, and the power-line noise to the MA. On the other hand, LVMLSS is not affected by the imbalance between PMOS and NMOS.

TABLE I
FEATURES OF THE TCAM TEST CHIP B

Technology	65nm high-performance CMOS process with 1-poly, 7-Cu metals and AL-top-metal
Supply voltage	1.0 V (core), 2.5V (I/O)
Total capacity	18 Mbit
Configuration	4.5 Mbit (64k entries $\times$ 72 bits) $\times$ 4 macros
Physical macro size	3.3 mm $\times$ 5.6 mm (18.5 mm ²) for 4.5Mb (area overhead of VDC is 5 %)
Physical bitcell area	1.69 μm^2
Search speed	From 50 MHz to 250 MHz
Power consumption	9.3 W for 18Mbit at 250 MHz, 1V, 25°C (including VDC power, which is 5 % of total power)

The influence of power noise is far less because LVMLSS is a differential-type MA and the reference voltage is generated at each cycle.

An 18-Mb full-TCAM test chip B with LVMLSS and PE is fabricated in a 65-nm CMOS process. The photographs of the TCAM bitcell and the test chip B are shown in Figs. 16 and 17. The cell size is 1.69 μm^2 for 65-nm technology, the size of which is just twice that of the 8T-SRAM bitcell. The die size is 99.06 mm² and the configuration is 256 k-entries $\times$ 72 b. The area overhead of LVMLSS compared with clocked inverter type MA is 11.9%. The shmoo plot of supply voltage vs. CLK cycle rate is shown in Fig. 18.

The power consumption of full parallel search is 9.3 W at 250 MHz. This figure includes the power consumption of VDC. The ratio of VDC power to the total power is 5%. A 250-MHz full parallel search operation is achieved at 1.0 V supply voltage.

Fig. 18 shows the power comparison per 1 Mbit. The power consumption of this work is reduced by 66% compared with previous work [3] which is chosen from large capacity TCAMs

TABLE II
PERFORMANCE COMPARISON

	[3] 2003	[9] 2004	[7] 2011	[15] 2013	This work
Key technology	CIMDC	ML with 4 segment	butterfly ML	early-predict late-correct	LVMLSS
Memory Size [Mb] Configuration	9 8 x 16k x 72	9 8 x 16k x 72	0.036 256 x 144	1.25 2k x 640	18 4 x 64k x 72
Technology [nm]	130 CMOS	130 CMOS	65 CMOS	32 SOI	65 CMOS
Supply Voltage [V]	1.5/1.2 *1	1.2	1.0	0.95	1.0
Max Frequency [MHz]	200	166	400	1000	250
TCAM cell size [μm^2]	6.73	7.1	7.05	N/A	1.69
1/density [mm ² /Mb]	19.4 *2	13	12	1.2	4.1
1/density (normalized to 65nm)	4.9	3.3	12	4.9	4.1
Energy/bit/search [fJ/bit/search]	5.60	2.87	0.165	0.58	1.98
Energy/bit/search (Normalized to 65nm)	2.1	1.0	0.165	1.31	1.98
FoM (E/bit/search) x (1/density)	9.8	3.2	2.0	6.4	8.1
Hierarchical ML	no	yes	yes	yes	no
Priority Encoder	yes	yes	no	no	yes

*1 Operating speed and power consumption in this table are measured at $V_{DD} = 1.5$ and 1.2 V, respectively.

*2 The core size is not written in [3], and this figure is calculated from CHIP size. Therefore, the figure is pessimistic.

3 For normalization for E/bit/search, the equation $E^ = E \times (65/\text{Technology}) \times (1.0/V_{DD})^2$ defined in [7] and (2) is used. For normalization for 1/density, the factor of $(65/\text{Technology})^2$ is multiplied.

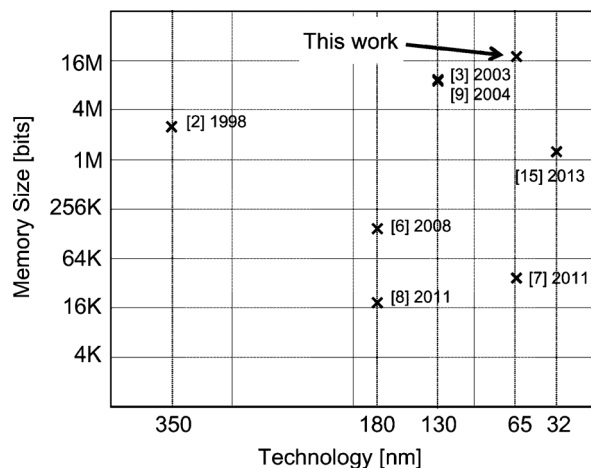


Fig. 20. TCAM capacity versus technology node. The memory size corresponds to 2x SRAM size.

with PE and without hierarchical (or segmented) ML configuration because the power of hierarchical ML depends on search data pattern. The energy consumption of this chip is 1.98 fJ/bit/search.

For further analysis, Fig. 19 also includes the power of our previous work in 90 nm (unpublished) and its normalized value, which uses a clocked inverter-type MA. From this comparison,

the power consumption of MA decreases from 150 mW (46%) to 50 mW (21%); however, the power of control signals increases from 50 mW (16%) to 70 mW (27%) because the number of control signal in LVMLSS MA is larger than that of a conventional clocked inverter-type MA.

Table I summarizes the feature of test chip B, and Fig. 20 shows the capacity of published CAM chips versus technology node. This work is the largest chip in the literature. Table II shows the performance comparison among those works of which TCAM have larger capacity than 1 Mb. As a top performance work on energy/bit/search, [7] is included. Those works which have hierarchical MLs are showing better energy performance. However, the values are awfully dependent on table data and key data patterns. The normalization on data pattern is not considered in the table.

VI. CONCLUSION

In this paper, an 18-Mb full ternary CAM with a low-voltage matchline sensing scheme (LVMLSS) has been designed and fabricated in 65-nm bulk CMOS process. LVMLSS has three key techniques: VDC, differential sense amplifier with matchline isolation, and reference voltage generation scheme. With these techniques, LVMLSS can reduce the dynamic power consumption of matchlines to 33% compared with conventional

one and realizes 42% faster matchline sensing. At 1.0 V typical supply voltage, 250 MHz search frequency is achieved. The power consumption of fully paralleled search operation at 250 MHz is 9.3 W, which is 66% smaller than previous work. This work has realized high speed, low power, and robust large scale TCAM by circuit level scheme. The combination of it with architectural and algorithmic level schemes can reduce the power further. We believe that this work will greatly contribute to reducing the power of network systems.

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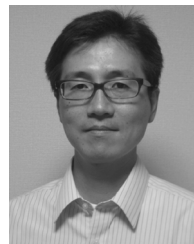
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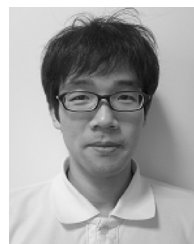
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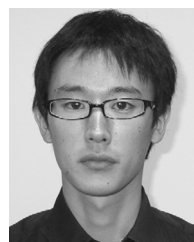
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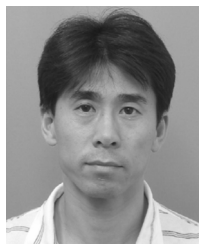
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